

Shubham Bhattacharya

AI Backend Engineer

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SUMMARY

AI Backend Engineer (BCA Final Year, Parul University) specializing in building production-grade LLM infrastructure — custom ReAct orchestration, multi-model fallback routing, Zod-validated tool registries, and prompt injection protection in TypeScript and Node.js. Proven track record of reducing inference costs by 30% and maintaining 99.9% API uptime through async queue design and model failover. Seeking an AI Backend Engineering internship where I can own agent infrastructure, optimize LLM serving costs, and ship reliable, observable systems.

SKILLS

Languages: TypeScript, JavaScript (Node.js), Python, SQL

Backend & API: Express.js, Next.js, FastAPI, REST API Design, Middleware Architecture, Rate Limiting, Auth (Clerk)

AI & LLM Infrastructure: Custom ReAct Loops, Tool Registry Design, Multi-Model Routing, Prompt Injection Protection, RAG, Vector Embeddings (pgvector, Xenova Transformers), tiktoken Cost Estimation, LLM Observability

Databases & Queues: PostgreSQL, Prisma ORM, Redis, BullMQ, MongoDB, pgvector

Observability & Safety: Zod Schema Validation, Sentry Error Tracking, Pino Structured Logging, OpenTelemetry

Testing & CI/CD: Vitest, Supertest, Playwright E2E, GitHub Actions, Automated Testing Pipelines

Infrastructure: Docker, AWS (ECS, S3, CloudWatch), Vercel, CI/CD Deployment

PROJECTS

Gigi AI Ops — Agentic E-commerce Backend

Node.js, TypeScript, Express, Groq SDK, PostgreSQL, Prisma, Redis, Zod, Sentry

- Architected a custom ReAct orchestration engine (no LangChain) processing 12,000+ daily order-tracking and refund requests autonomously, reducing manual support time by 85%
- Built a Zod-validated tool registry enforcing schema validation on every LLM output — 0 unvalidated tool calls reached the database across 10,000+ production executions
- Implemented multi-model fallback routing (Llama 3.3-70B to Llama 3.1-8B) via Groq SDK, maintaining 99.9% API uptime during provider outages; validated with 500+ stress tests at 0% failure rate
- Integrated tiktoken-based cost tracking; reduced average per-session LLM cost by 30% through intelligent model tiering (INR 0.04/session baseline)
- Deployed database-level audit logging identifying 2 query bottlenecks; cut p95 latency from 2.4s to 1.7s (30% improvement)
- Implemented Trusted Backend Context pattern for prompt injection protection — 200 adversarial test cases with 0% bypass rate

github.com/shubhambhattacharya-dev/Agentic_Project | agentic-project-drab.vercel.app

Medo Copilot — Multi-Provider AI Audit Platform

Next.js, TypeScript, Vercel AI SDK, Playwright, Clerk, Gemini, Groq, OpenRouter

- Designed a 3-tier LLM waterfall (Gemini 2.0 to Groq/Llama 4 Scout to OpenRouter/Claude) achieving 99.9% audit job completion rate with automatic provider failover
- Built hybrid scoring engine combining 60% AI-generated analysis with 40% deterministic Lighthouse and static code metrics — improved audit accuracy from 78% to 93% across 1,200+ production launches
- Developed automated static security scanner using Playwright visual regression, reducing false positive rate from 22% to 4%

github.com/shubhambhattacharya-dev/Medo_Copilot | medo-copilot.shubhambhattacharya.dev

DocNow — Telemedicine SaaS Platform

Next.js, TypeScript, Clerk, Prisma, PostgreSQL, Vonage WebRTC

- Built an AI symptom triage classifier that routes patients to correct specialists — reduced wrong-department bookings by 40% across 10,000+ appointments
- Developed full-stack telemedicine platform with WebRTC video calls handling 10,000+ sessions at 99.5% connection success rate
- Implemented multi-role payout engine (Patients, Doctors, Admins) with automated platform fee calculation, cutting manual reconciliation from 8 hours/week to 30 minutes/week

github.com/shubhambhattacharya-dev/health | docnow-zeta.vercel.app

EDUCATION

Bachelor of Computer Applications (BCA) — Parul University, Vadodara | 2022 – 2026

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Computer Networks, Operating Systems, Object-Oriented Programming